1 Geometry, Steady Kinetic Model, and Shear–Stress Closure

We consider a parallel-wall channel with wall-normal coordinate $z \in [-1, 1]$. Particle orientation is described by a single periodic angle $\theta \in [0, 2\pi]$. The (steady) reduced Smoluchowski equation under a unidirectional flow $u(z)\hat{x}$ reads

$$\sin\theta\, \partial_z \Psi(\theta, z) - S(z)\partial_\theta\big(\sin^2\theta\, \Psi(\theta, z)\big) = d_T \partial_{zz}\Psi(\theta, z) + d_R \partial_{\theta\theta}\Psi(\theta, z), \tag{1}$$

with no-flux boundary conditions at the walls,

$$\big(\sin\theta\, \Psi - d_T \partial_z \Psi\big)\big|_{z=\pm 1} = 0, \tag{2}$$

and periodicity in θ. The active shear stress is

$$\Sigma_{xz}^{(p)}(z) = \alpha \int_0^{2\pi} \cos\theta \sin\theta\, \Psi(\theta, z)\, d\theta, \tag{3}$$

and we close the mean-subtracted shear via

$$S(z) = -\Sigma_{xz}^{(p)}(z) + \frac{1}{2}\int_{-1}^{1} \Sigma_{xz}^{(p)}(\xi)\, d\xi, \qquad \int_{-1}^{1} S(z)\, dz = 0. \tag{4}$$

2 Finite-Dimensional Spaces and Expansions

Let $V_\theta = \mathrm{span}\{\Phi_k(\theta)\}_{k=1}^{K}$ be the Fourier angular basis on $[0, 2\pi)$:

$$\Phi_k(\theta) = \begin{cases} 1 & k = 1, \\ \cos(n\theta) & k = 2n,\ n \geq 1, \\ \sin(n\theta) & k = 2n+1,\ n \geq 1. \end{cases} \tag{5}$$

With $K = 2H + 1$ modes for maximum harmonic H.
Let $V_z = \mathrm{span}\{Z_m(z)\}_{m=0}^{M-1}$ be the **Chebyshev** spatial basis on $[-1, 1]$:

$$Z_m(z) = T_m(z), \qquad T_m(z) = \cos(m \arccos z), \tag{6}$$

where T_m is the Chebyshev polynomial of the first kind.
Seek approximations:

$$\Psi_{K,M}(\theta, z) = \sum_{k=1}^{K}\sum_{m=0}^{M-1} x_{km}\Phi_k(\theta) Z_m(z), \qquad S_M(z) = \sum_{m=0}^{M-1} s_m Z_m(z), \tag{7}$$

and test with $v(\theta, z) = \Phi_i(\theta) Z_n(z)$.

3 Weak Form

Integrate the θ-advection by parts in θ and the z-diffusion by parts in z. For all $i = 1, \ldots, K$ and $n = 0, \ldots, M-1$:

$$\iint \Phi_i Z_n \sin\theta\, \partial_z \Psi \, d\theta\, dz + \iint (\partial_\theta \Phi_i) Z_n S \sin^2\theta\, \Psi \, d\theta\, dz + d_T \iint (\partial_z Z_n)(\partial_z \Psi)\Phi_i \, d\theta\, dz + d_R \iint (\partial_\theta \Phi_i)(\partial_\theta \Psi) Z_n \, d\theta\, dz - d_T \int_0^{2\pi} \Phi_i Z_n \partial_z \Psi \Big|_{z=-1}^{z=+1} d\theta = 0, \quad (8)$$

with the boundary term enforced through the no-flux condition, projected against $\{\Phi_i\}$ at $z = \pm 1$.

4 Quadrature and One-Dimensional Building Blocks

Choose quadrature nodes and weights:

- **Angular**: $\{(\theta_j, w_j^\theta)\}_{j=1}^{Q_\theta}$ on $[0, 2\pi)$ (equispaced trapezoidal is spectral for Fourier).
- **Spatial**: $\{(z_q, w_q^z)\}_{q=0}^{Q_z-1}$ on $[-1, 1]$ using **Chebyshev–Gauss–Lobatto** nodes:

$$z_q = -\cos\left(\frac{\pi q}{Q_z - 1}\right), \qquad q = 0, 1, \ldots, Q_z - 1, \quad (9)$$

  with **Clenshaw–Curtis** quadrature weights w_q^z satisfying

$$\int_{-1}^{1} f(z)\, dz \approx \sum_{q=0}^{Q_z-1} w_q^z f(z_q). \quad (10)$$

Quadrature inner products:

$$\langle f, g \rangle_{Q_\theta} := \sum_{j=1}^{Q_\theta} w_j^\theta f(\theta_j) g(\theta_j), \qquad \langle f, g \rangle_{Q_z} := \sum_{q=0}^{Q_z-1} w_q^z f(z_q) g(z_q). \quad (11)$$

4.1 Angular Matrices

$$(M_\theta)_{ik} = \langle \Phi_i, \Phi_k \rangle_{Q_\theta}, \qquad (K_\theta)_{ik} = \langle \partial_\theta \Phi_i, \partial_\theta \Phi_k \rangle_{Q_\theta}, \quad (12)$$

$$(P_1)_{ik} = \langle \Phi_i, \sin\theta\, \Phi_k \rangle_{Q_\theta}, \qquad (P_2)_{ik} = \langle \Phi_i, \sin^2\theta\, \Phi_k \rangle_{Q_\theta}. \quad (13)$$

Jeffery kernel (row vector, length K):

$$(H_\theta)_k = \langle \cos\theta \sin\theta, \Phi_k \rangle_{Q_\theta}. \quad (14)$$

Discrete angular derivative: $(D_\theta)_{ik} = \langle \partial_\theta \Phi_i, \Phi_k \rangle_{Q_\theta}$.

4.2 Spatial Matrices (Chebyshev)

$$(M_z)_{nm} = \langle Z_n, Z_m \rangle_{Q_z} = \langle T_n, T_m \rangle_{Q_z}, \tag{15}$$
$$(K_z)_{nm} = \langle \partial_z Z_n, \partial_z Z_m \rangle_{Q_z} = \langle T_n', T_m' \rangle_{Q_z}, \tag{16}$$
$$(G_z)_{nm} = \langle Z_n, \partial_z Z_m \rangle_{Q_z} = \langle T_n, T_m' \rangle_{Q_z}. \tag{17}$$

Chebyshev boundary evaluation vectors (analytic formulas):

$$E_m^+ := T_m(+1) = 1, \tag{18}$$
$$E_m^- := T_m(-1) = (-1)^m, \tag{19}$$
$$D_m^+ := T_m'(+1) = m^2, \tag{20}$$
$$D_m^- := T_m'(-1) = (-1)^{m+1} m^2. \tag{21}$$

Integration vector:

$$b_m = \int_{-1}^{1} T_m(z)\, dz = \begin{cases} 2 & m = 0, \\ 0 & m \text{ odd}, \\ \dfrac{2}{1-m^2} & m \text{ even},\ m \geq 2. \end{cases} \tag{22}$$

Constant function expansion: $1 = \sum_m (c_z)_m T_m(z)$, so $(c_z)_0 = 1$ and $(c_z)_m = 0$ for $m \geq 1$.

4.3 Triple-Product Matrices

For the Jeffery term, we need triple products:

$$C_{npm}^{(z)} = \langle T_n, T_p \cdot T_m \rangle_{Q_z}. \tag{23}$$

Using the Chebyshev product identity $T_p T_m = \frac{1}{2}(T_{p+m} + T_{|p-m|})$, these satisfy selection rules.

5 Tensor-Product Assembly of the Interior Operator

Stack unknowns $\mathbf{x} \in \mathbb{R}^{KM}$ in θ-major order. Kronecker blocks:

$$\mathbf{M} = M_\theta \otimes M_z, \qquad \mathbf{K}_\theta = K_\theta \otimes M_z, \qquad \mathbf{K}_z = M_\theta \otimes K_z. \tag{24}$$

z**-advection**: From $\iint v \sin\theta\, \partial_z \Psi$:

$$A_z = P_1 \otimes G_z. \tag{25}$$

Jeffery term (integrated by parts in θ): With $J_\theta := D_\theta P_2$ and the spatial multiplication by $S(z)$,

$$(T_z(S))_{nm} = \langle Z_n, S(z) Z_m \rangle_{Q_z} = \sum_{p=0}^{M-1} s_p \langle T_n, T_p T_m \rangle_{Q_z} = \sum_{p=0}^{M-1} s_p C_{npm}^{(z)}, \tag{26}$$

the Jeffery contribution is

$$J(S) = J_\theta \otimes T_z(S), \tag{27}$$

linear in $\mathbf{s} = (s_0, \ldots, s_{M-1})^\top$.

6 Shear Map $\mathbf{s} = \mathbf{S}\mathbf{x}$ from the Closure

From the shear-stress closure:

$$\mathbf{a}^{(\Sigma)} = \alpha (H_\theta \otimes I_M)\mathbf{x} =: \alpha H \mathbf{x} \in \mathbb{R}^M, \tag{28}$$

where $H = H_\theta \otimes I_M$.
With $\mathbf{c}_z \in \mathbb{R}^M$ the coefficients of the constant function ($c_z = (1, 0, \ldots, 0)^\top$ for Chebyshev) and $b_m = \int_{-1}^{1} T_m \, dz$:

$$\mu = \int_{-1}^{1} \Sigma_{xz}^{(p)} \, dz = \mathbf{b}^\top \mathbf{a}^{(\Sigma)}, \tag{29}$$

$$\mathbf{s} = -\mathbf{a}^{(\Sigma)} + \frac{1}{2}\mu \, \mathbf{c}_z = \left(-\alpha H + \frac{1}{2} \mathbf{c}_z \mathbf{b}^\top \alpha H \right) \mathbf{x} =: \mathbf{S}\mathbf{x}. \tag{30}$$

7 Projected Wall Boundary Conditions and Normalization

Project the no-flux condition against $\{\Phi_i\}$ at $z = \pm 1$:

$$\langle \Phi_i, \sin\theta \, \Psi(\cdot, \pm 1) \rangle_{Q_\theta} - d_T \langle \Phi_i, \partial_z \Psi(\cdot, \pm 1) \rangle_{Q_\theta} = 0, \quad i = 1, \ldots, K. \tag{31}$$

This yields two K-row linear constraints on $\mathbf{x}$:

$$C_\pm \mathbf{x} = 0, \qquad C_\pm = (P_1 \otimes E^\pm) - d_T (M_\theta \otimes D^\pm). \tag{32}$$

Add one linear normalization (e.g., $\iint \Psi \, d\theta \, dz = 2$) and stack all as:

$$C\mathbf{x} = \mathbf{d}, \qquad \mathbf{d} = (0, \ldots, 0, 2)^\top \in \mathbb{R}^{2K+1}, \qquad C \in \mathbb{R}^{(2K+1) \times KM}. \tag{33}$$

8 Discrete Nonlinear System (KKT Form)

Define the interior residual:

$$F_{\text{int}}(\mathbf{x}) = A_z\mathbf{x} + J(\mathbf{S}\mathbf{x})\mathbf{x} + d_T\mathbf{K}_z\mathbf{x} + d_R\mathbf{K}_\theta\mathbf{x}. \tag{34}$$

Enforce constraints with multipliers $\boldsymbol{\tau}$:

$$\begin{bmatrix} A_z + d_T\mathbf{K}_z + d_R\mathbf{K}_\theta + J(\mathbf{S}\mathbf{x}) & C^\top \\ C & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \boldsymbol{\tau} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{d} \end{bmatrix}. \tag{35}$$

Because $J(\mathbf{S}\mathbf{x})$ depends linearly on $\mathbf{x}$ and multiplies $\mathbf{x}$, this system is bilinear in $\mathbf{x}$.
Picard with under-relaxation: Define a fixed-point map $T(\mathbf{x})$: freeze S at $\mathbf{S}\mathbf{x}$, solve the linear KKT for $\mathbf{x}^{\text{new}}$, then apply under-relaxation:

$$\mathbf{x}^{(k+1)} = (1-\omega)\mathbf{x}^{(k)} + \omega\,\mathbf{x}^{\text{new}}. \tag{36}$$

9 Quadrature, De-aliasing, and Accuracy

Angular: For Fourier, equispaced trapezoidal with $Q_\theta \geq 2K$ integrates products spectrally; use $Q_\theta \approx 3K$ (the 3/2-rule) for the Jeffery term.
Spatial: For Chebyshev polynomials on Gauss–Lobatto nodes, Clenshaw–Curtis quadrature is exact for polynomials up to degree $2(Q_z - 1)$. Choose Q_z large enough for triple products; for degree M, $Q_z \gtrsim 2M$ is safe.

10 Chebyshev-Specific Implementation Notes

10.1 Clenshaw–Curtis Weights

For Chebyshev–Gauss–Lobatto nodes $z_q = -\cos(\pi q/(Q_z - 1))$, the Clenshaw–Curtis weights are:

$$w_q = \frac{c_q}{Q_z - 1} \sum_{k=0,\, k \text{ even}}^{Q_z-1} \frac{b_k}{1-k^2} \cos\left(\frac{\pi q k}{Q_z - 1}\right), \tag{37}$$

where $c_0 = c_{Q_z-1} = 1/2$ and $c_q = 1$ otherwise, and $b_0 = b_{Q_z-1} = 1$, $b_k = 2$ for $0 < k < Q_z - 1$.

10.2 Chebyshev Derivative

The derivative of $T_m(z)$ is:

$$T_m'(z) = mU_{m-1}(z), \tag{38}$$

where U_{m-1} is the Chebyshev polynomial of the second kind. For evaluation at quadrature points, use the recurrence or compute directly from $T_m'(z) = m\sin(m\arccos z)/\sqrt{1-z^2}$ (handling endpoints specially).
At the boundaries:

$$T_m'(\pm 1) = (\pm 1)^{m+1} m^2. \tag{39}$$

10.3 Advantages of Chebyshev Basis

- **Direct compatibility with Dedalus**: The EVP solver uses Chebyshev bases, so Galerkin coefficients transfer directly without interpolation.
- **FFT-based transforms**: Chebyshev coefficients can be computed via DCT in $O(M \log M)$ time.
- **Simple boundary values**: $T_m(\pm 1)$ and $T'_m(\pm 1)$ have closed-form expressions.
- **Optimal approximation**: Chebyshev polynomials minimize the maximum error for polynomial interpolation.

11 Minimal Assembly Checklist

1. Build 1D angular blocks M_θ, K_θ, P_1, P_2, D_θ, H_θ by quadrature.
2. Build 1D spatial blocks M_z, K_z, G_z, $E^\pm$, $D^\pm$ and triple products $C^{(z)}_{npm}$ using Chebyshev polynomials on Gauss–Lobatto nodes with Clenshaw–Curtis weights.
3. Set $A_z = P_1 \otimes G_z$, $\mathbf{K}_\theta = K_\theta \otimes M_z$, $\mathbf{K}_z = M_\theta \otimes K_z$.
4. Given $\mathbf{x}$, compute $\mathbf{s} = \mathbf{S}\mathbf{x}$; form $T_z(S) = \sum_p s_p T_z^{(p)}$.
5. Form $J(S) = J_\theta \otimes T_z(S)$ with $J_\theta = D_\theta P_2$.
6. Assemble wall constraints $C_\pm$ and append normalization to get C.
7. Solve the KKT system via Picard iteration with under-relaxation.